

# SIADS 696 Milestone II Project Report

## Galaxy Classification

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### Introduction

Space is cool. Earth is inside the Milky Way Galaxy, classified as a barred spiral galaxy. The closest galaxy to us is the Andromeda Galaxy, also a barred spiral galaxy. Astronomers predict that the two galaxies will collide in about 4.5 billion years. What does it look like when two barred spiral galaxies collide? You get a giant elliptical galaxy in the end (Staff, 2020). This isn't just a fun fact, classifying galaxies by shape is how astrophysicists begin to understand the life cycle of the galaxy; it allows them to reconstruct the history of galactic mergers and understand how matter redistributes over time.

Developed in 1926 by Edwin Hubble, the Hubble Tuning Fork diagram is used to group galaxies by shape.

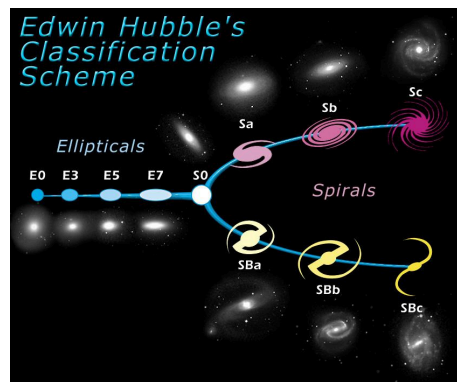


Figure 1. Edwin Hubble's Classification Scheme (image credit: <https://esahubble.org/images/heic9902o/>)

The diagram is divided into two shapes – elliptical and spiral. The scheme is largely considered too simple today, but its basic ideas still hold: start with classifying based on the basic shape from the tuning fork to understand where in the lifecycle a galaxy is, then conduct more specific research and observation. Young galaxies are usually spiral in shape, old galaxies are usually elliptical; galaxies that don't fit into the spiral or elliptical shapes are usually irregular because they are merging.

Remember at the beginning of Star Wars where the text reads “in a galaxy far far away...” Well, all galaxies are far far away. The closest galaxy to us, the previously mentioned Andromeda galaxy, is approximately 2.5 million light-years away. Although not impossible, it's pretty hard to see things that far away with the naked eye. Additionally, the shape of a galaxy is dictated by things we can't see. It's widely accepted that most galaxies have supermassive black holes at their center and that the way a galaxy holds its shape has to do with the distribution of dark matter surrounding it. Although the terms are vague, “black holes” and “dark matter” invoke the idea that these are things we can't casually observe.

Historically, astronomers classified galaxies by hand. However, modern space telescopes and sky surveys capture images of billions of galaxies. Human eyes simply cannot process this much data. Automated, highly accurate machine learning models are needed to analyze the millions of terabytes of cosmic data coming from these telescopes.

One issue we're trying to solve is how to balance a model with the different photometric filters that get used on telescopes. A very important consideration is how different the filters are on land-based telescopes from space-based telescopes. Land-based telescopes need to account for how light passes through the atmosphere; space-based telescopes do not. But it's not as simple as just making a bunch of space-based telescopes to do all the analysis. First of all, it's expensive to go to space. Second of all, maintenance is *difficult*. The challenge we wanted to explore is how to use the institutional knowledge built from the land-based telescope datasets and apply it to the newer datasets generated by space-based telescopes.

This gap is about domain adaptation. Models trained on land-based data will likely fail when applied to space-based data because of the photometric filter differences. Developing a model that can seamlessly bridge the gap is an essential step to unlocking the full potential of next-generation space telescopes without just tossing out the decades (nearly a century) worth of accumulated land-based knowledge. The entirety of that project is beyond our scope, but it's what we started to peek into.

### *Motivation*

We were motivated to explore this topic because of our interest in astronomy and astrophysics. One group member, Nancy, had started a galaxy classification back in 2017 as part of a summer program at the American Museum of Natural History (AMNH) in New York. That classification project was focused on using spectra to further identify what is known as E + A (Elliptical + A-type stars) galaxies—post-starburst galaxies that have experienced a sudden quenching phase. That initial interest combined with the launch of the James Webb Space Telescope and resurgence of GalaxyZoo project that uses citizen astronomers to classify new images fueled our interest in this project. Space is so vast, beautiful, and complicated we thought it would be cool to bring machine learning techniques to traditional classification methods; to see if citizen data scientists could reach similar results as citizen astronomers.

### *Summary of Our Methods*

In order to maximize our exploration of the data and better understand how various machine learning methods could be used to classify or group the data, we structured our project to include both supervised and unsupervised models that worked by building off of each other. Our Supervised portion was where we created our first classification models, like the logistic regression and random forest models. These models did not perform very well on the first round. Using our unsupervised learning methods, we were able to find some underlying patterns in the data, however, as we are not astrophysicists, a full understanding of what these patterns could mean would require some additional knowledge. The clusters that were found in the KMeans model were used as features in our supervised models to see if the classifier's accuracy would improve. In the case of logistic regression or random forest models, the accuracy did not improve much. Then we extracted CNN embedding and used those in a random forest model and a logistic regression model, which did lead to improved performance. Our CNN model was already our best performing supervised learning model, even if there would be no improvement from feature engineering, we would have an accurate classifier.

### *Findings*

Through our supervised and unsupervised models, we found that galaxies are hard to classify using just morphological data; the images are paramount. Through our clustering techniques, we've determined that most galaxies are more similar than dissimilar with the exception of a few outliers. A lot of the data points fell closely together, which led to trouble producing multiple clusters.

## Related Work

Understanding and classifying galaxies has been a focus of many works in astronomy. One large-scale morphological classification study is the Galaxy zoo Project (Lintott et al., 2008). This project uses human volunteers to categorize millions of objects. This project has done so much for the field, however, relying solely on human eyes takes many years, can introduce bias, and lacks computational accuracy.

Projects like Ghaderi, Alipour, and Safari's Galaxy Morphological Classification with Zernike Moments and Machine Learning Approaches (Ghaderi et al., 2025) use automated morphological classification. Their model used traditional machine learning methods to capture geometric properties of galaxy images.

Our project used CNNs that were ultimately the most accurate model, and projects like Huertas-Company's Deep Learning (Huertas Company et al., 2019) approach set an excellent precedent. This study utilized deep Convolutional Neural Networks in a fully supervised way by mapping raw pixel grids directly to morphology classes.

Our project built directly on the ideas presented in these journals while introducing new modifications. Unlike the standard supervised approaches, or only using unsupervised modeling, we aimed to combine the two methods. We used unsupervised models to cluster our data and find underlying patterns, and we used supervised learning models trained on the labeled morphological data and images to classify the galaxies. Where we diverge from these studies is in the use of our findings in our unsupervised models as features in our supervised learning models.

## Data Sources

Our two data sources were the [GalaxiesML](#) dataset and the JWST (James Webb Space Telescope) portion of the [Multimodal Universe](#) dataset.

The GalaxiesML dataset can be downloaded from <https://zenodo.org/records/13878122>. The downloaded files are images in HDF5 format and CSVs with metadata. The dataset is split into training, testing, and validation sets. The most important variables were 'image', 'specz\_redshift', and '{band}\_sersic\_index' for each band of photometric filters in the dataset: g, r, i, z, and y. We used all 286,401 records from the dataset. The data is from a public data release in August 2021.

The Multimodal Universe dataset is hosted at the Flatiron Institute and can be downloaded from <https://users.flatironinstitute.org/~polymathic/data/MultimodalUniverse>. The downloaded files are HDF5 files. There are 645 columns in this dataset; the most important for us was 'image\_flux' which was the actual image. Of the 300,000 available images in the dataset, split up by different sections of the sky, we used the portion from the PRIMER (Public Release IMaging for Extragalactic Research) survey from the healpix=1703 section of the sky; the dataset contained 51,058 images.

## Feature Engineering

The GalaxiesML dataset is designed for machine learning, and provides training, validation, and test splits, making feature engineering a very minimal task. There were no missing data points or otherwise noisy data to deal with before we could begin processing.

The main step we took in order to use the dataset for supervised learning was to create labels from the Sérsic index feature contained in the morphological data. Sérsic index is a value that measures how the intensity of light varies from the center of the galaxy. In astrophysics, the correlation of Sérsic index with galaxy morphology is sometimes used to determine the Hubble type of distant galaxies (*Sérsic Profile*, n.d.). We took advantage of this rule of thumb to create masks for the data to assign labels. From there we dropped the Sérsic index from all datasets used for modeling.

Our first run of the logistic regression model used the magnitude of the central galaxy in the filter for each band: `g_cmodel_mag`, `r_cmodel_mag`, `i_cmodel_mag`, `z_cmodel_mag`, and `y_cmodel_mag`. Our second run used the ellipticity for each photometric band as features. Our random forest model also used the `c_mag` values while the CNN model used the image as its only feature. We used ‘image\_flux’ and the ellipticity features of each photometric band for our unsupervised learning DBSCAN model, where ‘image\_flux’ is the image in the JWST dataset. Features used for supervised and unsupervised were essentially the same, just named slightly differently. We chose these features because they had the most potential to be good predictors for galaxy shape.

With help from some astrophysics journal articles *The morphological mix of dwarf galaxies in the nearby Universe* (Lazar et al., 2024) and *Galaxy populations in the most distant SPT-SZ clusters* (Strazzullo et al., 2023), we created 5 common labels that are inline with the general Hubble Tuning Fork classification: ETG/ Classical Bulge, Giant Elliptical / BCG, Irregular, LTG / Disk, and Pseudobulge / Transition.

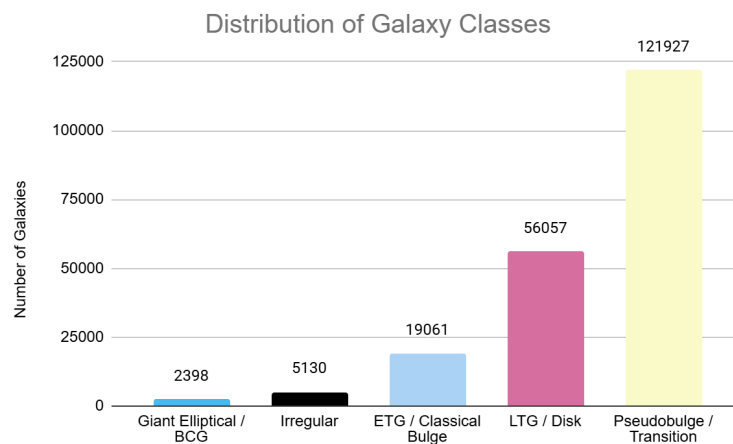
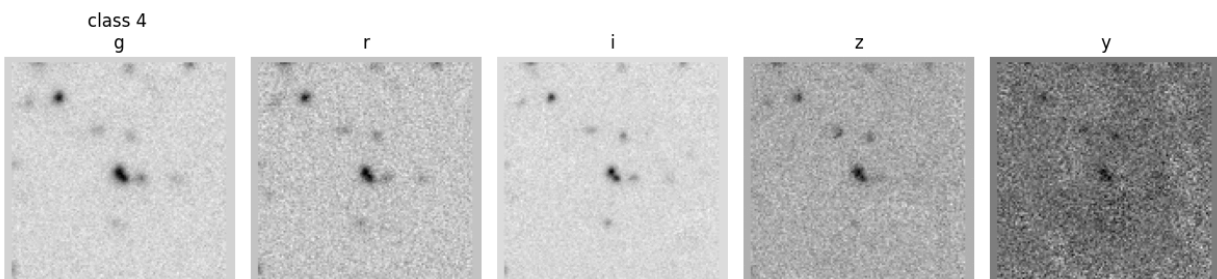


Figure 2: Distribution of Galaxy Classes in the GalaxyML training split. Colors of each distribution bar roughly correspond to the colors in Figure 1.

The images from the dataset do not look like the beautiful images of space you are used to seeing from NASA, which is part of why deciphering the images is so paramount to astrophysics work.



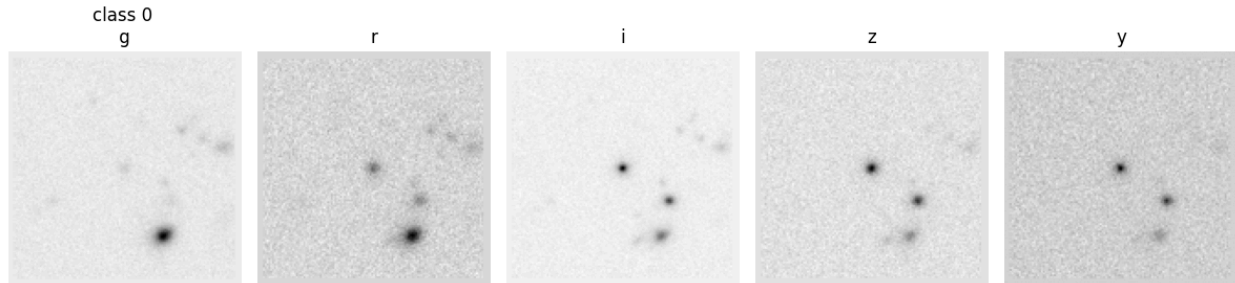


Figure 3: Top—image of a class 4 (Irregular) galaxy, one image per photometric filter.  
Bottom—image of a class 0 (LTG / Disk) galaxy, one image per photometric filter.

## Part A. Supervised Learning

### *Methods Description*

Our supervised learning portion was a two-pronged approach, though admittedly it didn't start out that way. We started with a logistic regression model on the GalaxyML dataset to define an interpretable baseline for tabular morphological data. The mathematical simplicity of a logistic regression serves as a critical point of comparison. We chose the logistical regression in order to evaluate if more complex models, like the random forest, and the deep learning architecture, the CNN, yield a justifiable increase in classification accuracy.

After the linear baseline, we implemented a random forest model on the tabular morphological data. Since galaxies tend to be complicated and morphological data typically displays nonlinear structures and interdependent features, the linear boundaries found with the logistic regression weren't going to be robust enough. The random forest model would address the complex morphological data with very minimal manual feature engineering. This was important to us since we are not ourselves astrophysicists and therefore needed the assist when it came to said feature engineering. Random forests also provide resistance to noise and outliers, which are important due to the size of the dataset and how varied galaxies can look when they are within the irregular class.

As a third method, and because we had image data, we built a CNN to leverage the spatial structure and multiband image information that isn't available in the tabular data. Most CNNs are built using three bands, representing the Red Green Blue (RGB) color bands present in most color images; there is a deep history of the RGB color scale and its persistence. However, images taken of space reach beyond the visible light spectrum into other wavelengths of the electromagnetic spectrum. For the GalaxyML data, we adjusted the standard ResNet18 CNN model to 5 bands to account for the g (green), r (red), i (infrared), z (near-infrared), and y (deep near-infrared) photometric filters used on the telescope capturing the images. The CNN was trained on just the images and the labels without any morphological data.

During our proposal stage, we had intended to use morphological data alongside image data with the pretense that the model would be stronger. However, the initial evaluation of the logistic regression and random forest compared to that of the CNN is what led us to the two-pronged approach.

After the logistic regression and random forest yielded low values for every evaluation metric (precision, recall, f1-score), while the CNN yielded relatively high values, we decided to extract the CNN embeddings and then train the logistic regression and random forest on those embeddings. This yielded much better results.

### Supervised Evaluation

To evaluate our supervised learning models, we prioritized macro F1 because of the class imbalance, as shown in Figure 2. We also looked at weighted F1 to reflect aggregate performance with class-frequency weighting, minority-class recall to monitor failure on rare classes, and accuracy as a familiar secondary metric.

Initial exploration of the logistic regression model involved switching features. Our first run used the magnitude of the central galaxy in the filter for each band: `g_cmodel_mag`, `r_cmodel_mag`, `i_cmodel_mag`, `z_cmodel_mag`, and `y_cmodel_mag`. We chose these features because the magnitude of the galaxy *should* lead to reasonable predictions about its class. This model resulted in an accuracy value of 0.23 and F1-scores between 0.02 to 0.31.

We then switched to using the ellipticity for each photometric band as features because a measure of ellipticity should also be a good predictor of galaxy class, specifically if it is an elliptical galaxy or not. The model resulted in an accuracy of 0.19 with F1-scores between 0.02 and 0.27.

|   | precision | recall | f1-score | support |   | precision | recall | f1-score | support |
|---|-----------|--------|----------|---------|---|-----------|--------|----------|---------|
| 0 | 0.43      | 0.24   | 0.31     | 11211   | 0 | 0.37      | 0.21   | 0.27     | 11211   |
| 1 | 0.72      | 0.14   | 0.23     | 24386   | 1 | 0.65      | 0.1    | 0.18     | 24386   |
| 2 | 0.17      | 0.72   | 0.27     | 3812    | 2 | 0.16      | 0.63   | 0.25     | 3812    |
| 3 | 0.01      | 0.18   | 0.02     | 480     | 3 | 0.01      | 0.015  | 0.02     | 480     |
| 4 | 0.1       | 0.72   | 0.17     | 1026    | 4 | 0.06      | 0.51   | 0.11     | 1026    |

Figure 4: Left—Precision, Recall, F1-score and support values for predicted classes in the logistic regression using central magnitude as features.  
Right—Precision, Recall, F1-score, and support values for predicted classes in logistic regression using ellipticity as features.

We took a similar approach for the random forest model. With such little difference between the ellipticity and central magnitude features, we decided to use the central magnitude features for the random forest. The random forest performed better with an accuracy of 0.44 and F1 scores ranging from 0.04 to 0.53, but these results were still not ideal.

Our next step was hyperparameter tuning. We used 5-fold stratified cross-validation and optimized for macro F1 for both the logistic regression and the random forest. The results for the best logistic regression was a macro F1 score of 0.2241. For the random forest the best model returned a macro F1 score of 0.3702.

The CNN was trained using only the ‘image’ feature and the labels. Eventually we received the results of a loss value of 0.2382 with an accuracy score of 0.9030. From there we performed CNN hyperparameter tuning where we defined a grid of parameter combinations of learning rates, weight decay, batch size, epoch, and optional drop out. We used the validation metric of macro F1 again for early stopping and model choice.

After hyperparameter tuning, our best performing CNN was with a learning rate of 0.001, a batch size of 32, a weight decay of 0.001 using the ‘adamw’ optimizer with 10 epochs and a patience level set to 3. The CNN was evaluated with a stratified 5-fold cross-validation using Macro F1 as the primary metric.

| K Folds | Macro F1 Mean | Macro F1 std | Weighted F1 mean | Weighted F1 std | Accuracy Mean | Accuracy std | Minority Recall Mean | Minority Recall std |
|---------|---------------|--------------|------------------|-----------------|---------------|--------------|----------------------|---------------------|
| 5       | 0.7782        | 0.0090       | 0.8972           | 0.0055          | 0.8978        | 0.0062       | 0.5749               | 0.0613              |

Figure 5: CNN 5-fold cross-validation summary

Seeing the large difference in metrics between the CNN and the logistic regression and random forest models, we decided to extract the CNN embeddings and use those to train the logistic regression and random forest models. Since we had saved the best model during hyperparameter tuning, we were able to revert to the best checkpoint and extract the embeddings from that model.

|   | precision | recall | f1-score | support |   | precision | recall | f1-score | support |
|---|-----------|--------|----------|---------|---|-----------|--------|----------|---------|
| 0 | 0.89      | 0.88   | 0.88     | 11069   | 0 | 0.9       | 0.9    | 0.9      | 11069   |
| 1 | 0.97      | 0.89   | 0.93     | 24465   | 1 | 0.94      | 0.95   | 0.94     | 24465   |
| 2 | 0.75      | 0.86   | 0.8      | 3906    | 2 | 0.86      | 0.81   | 0.83     | 3906    |
| 3 | 0.38      | 0.82   | 0.52     | 466     | 3 | 0.75      | 0.59   | 0.66     | 466     |
| 4 | 0.45      | 0.83   | 0.59     | 1008    | 4 | 0.7       | 0.6    | 0.64     | 1008    |

Figure 6: Left—Precision, recall, F1, and support values of logistic regression trained on CNN embeddings.  
Right—Precision, recall, F1 score, and support values of random forest trained on CNN embeddings.

Both logistic regression and random forest performance improved using the CNN embeddings. Since the CNN embeddings are learned features, the models get cleaner inputs allowing them to separate the classes well. Therefore models trained on the CNN embeddings likely improved because the learned embedding space is more informative while also reducing pixel noise, making class separation easier. The random forest actually outperformed the original CNN.

| Model                     | Feature Representation                                       | Best Hyperparameters                                                                                                   | Macro F1 | Weighted F1 | Minority-Class Recall | Notes                                                        |
|---------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------|-------------|-----------------------|--------------------------------------------------------------|
| CNN (ResNet18, 5-channel) | 5-band image tensor (g, r, i, z, y); spatial pixel structure | lr: 0.001, 'batch_size': 32, 'weight_decay': 0.0001, 'optimizer': 'adamw', 'epochs': 12, 'patience': 5                 | 0.7921   | 0.9045      | 0.6102                | Best run: 20260619-145503_lr0.001_bs32_wd0.0001   grid_run_2 |
| Logistic Regression       | Tabular photometric / morphology summary features            | model__C': 10.0, 'model__class_weight': None, 'model__max_iter': 3000, 'model__penalty': 'l1', 'model__solver': 'saga' | 0.2241   | 0.5413      | 0                     | CV macro F1=0.2222                                           |

|                                      |                                                   |                                                                                                                                         |        |        |        |                                                  |
|--------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------------------------------------------------|
| Random Forest                        | Tabular photometric / morphology summary features | class_weight':<br>'balanced_subsample',<br>'max_depth': 20,<br>'max_features': 'sqrt',<br>'min_samples_leaf': 5,<br>'n_estimators': 500 | 0.3702 | 0.6088 | 0.0146 | CV macro F1=0.3763                               |
| Logistic Regression (CNN embeddings) | Fixed CNN embedding vector                        | solver=saga,<br>class_weight=balanced,<br>max_iter=3000                                                                                 | 0.7449 | 0.8911 | 0.8219 | Embedding-space classifier (test);<br>acc=0.8841 |
| Random Forest (CNN embeddings)       | Fixed CNN embedding vector                        | n_estimators=500,<br>class_weight=balanced_s<br>ubsample,<br>random_state=42                                                            | 0.7959 | 0.9109 | 0.5879 | Embedding-space classifier (test);<br>acc=0.9120 |

Figure 7: Model Comparison

*In Depth Evaluation*

Our best performing model was the random forest trained on CNN embeddings.

*Feature Importance and Ablation Analysis*

Feature importance for the random forest trained on the CNN embeddings shows a highly uneven distribution. The most informative dimensions have importance on the order of  $10^{-2}$ , whereas many other dimensions are near  $10^{-6}$  to  $10^{-8}$ . This shows that class discrimination is concentrated in a relatively small subset of embedding dimensions while many only minimally contribute. Future work could evaluate dimensionality reduction to reduce model complexity with minimal performance loss.

*Sensitivity Analysis*

We varied the n-estimators, max\_depth, and min\_samples\_leaf hyperparameters around the embedding-based baseline to measure the robustness of Macro F1, weighted F1, and minority recall.

| Num Estimators | Max Depth | Min Samples Leaf | Macro F1 | Weighted F1 | Minority Recall |
|----------------|-----------|------------------|----------|-------------|-----------------|
| 800            |           | 2                | 0.7991   | 0.9113      | 0.6202          |
| 500            | 40        | 2                | 0.7991   | 0.9114      | 0.6202          |
| 800            | 40        | 2                | 0.7991   | 0.9113      | 0.6266          |
| 500            |           | 2                | 0.7986   | 0.9112      | 0.6180          |
| 300            |           | 2                | 0.7985   | 0.9110      | 0.6223          |
| 300            |           | 1                | 0.7982   | 0.9111      | 0.5944          |
| 300            | 40        | 2                | 0.7980   | 0.9115      | 0.6180          |
| 800            | 40        | 1                | 0.7969   | 0.9111      | 0.6030          |
| 800            |           | 1                | 0.7964   | 0.9106      | 0.5901          |
| 500            |           | 1                | 0.7959   | 0.9109      | 0.5880          |

Figure 8: Random Forest Sensitivity Results

The range of Macro F1 scores from 0.7919 to 0.7991 indicated the random forest is not incredibly sensitive to hyperparameter tuning.

### Tradeoffs

We compared the CNN, the logistic regression on CNN embeddings, and the random forest on CNN embeddings using Macro F1, weighted F1, and minority recall.

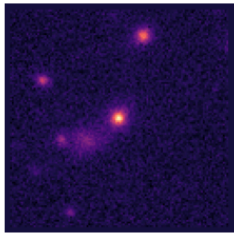
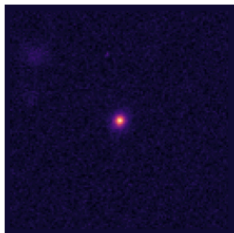
| Model                                | Macro F1 | Weighted F1 | Minority Recall | Evaluation Protocol        |
|--------------------------------------|----------|-------------|-----------------|----------------------------|
| CNN (ResNet18, 5-channel)            | 0.7921   | 0.9045      | 0.6102          | Best CNN run (validation)  |
| Logistic Regression (CNN embeddings) | 0.7449   | 0.8911      | 0.8219          | Holdout on test embeddings |
| Random Forest (CNN embeddings)       | 0.7959   | 0.9109      | 0.5880          | Holdout on test embeddings |

Figure 9: Trade Off Summary Table

We indicated the random forest trained on CNN embedding was our best performing model based on Macro F1 score. However, the logistic regression did have a better minority recall score. This does have implications that we are deciding to forgo the model that preserves the worst-served class. The random forest on embeddings delivers the stronger overall aggregate performance, reflecting a fairness versus overall performance tradeoff.

### Failure Analysis

Three examples where prediction failed:

| sample_index | true_label | pred_label | true_class_name          | pred_class_name          | Image Sample (y band)                                                                 |
|--------------|------------|------------|--------------------------|--------------------------|---------------------------------------------------------------------------------------|
| 24           | 1          | 0          | Pseudobulge / Transition | LTG / Disk               |  |
| 35           | 2          | 1          | ETG / Classical Bulge    | Pseudobulge / Transition |  |

|    |   |   |            |                          |                                                                                     |
|----|---|---|------------|--------------------------|-------------------------------------------------------------------------------------|
| 48 | 0 | 1 | LTG / Disk | Pseudobulge / Transition |  |
|----|---|---|------------|--------------------------|-------------------------------------------------------------------------------------|

Figure 10: Three failed prediction records

The failure of sample index 24 is likely due to noise in the image. There appear to be more than one galaxy or other celestial entity in the image, especially close to the center of the image, which can cause model failure. The image is also generally noisy, which is more obvious in the other photometric bands.

The failure in sample 35 is likely due to boundary failure. The pseudobulge / transition class and ETG / classical bulge class are next to each other in the evolution of galaxy classes. There is a thin margin between these two classes based on the sércic index, with pseudobulge ranging from 1.0 up to 2.5 and ETG/classical bulge starting at 2.5 and ranging to 4.0. If the galaxy that sits at sample index 35 had a sércic index close to 2.5, it could be victim to boundary ambiguity.

The failure in sample 48 is likely due to confusion between similarly visual classes. The LTG / Disk class and Pseudobulge / Transition class are visually similar, and even though to the naked eye this image appears to be a disk, the class imbalance between the two classes combined with the visual similarities likely led to the misclassification.

To mitigate the failures in the future, we could implement reweighting during training to improve minority class recall, allowing them to contribute more strongly and encourage the model to recover rare classes rather than optimizing for majority-class accuracy. We could also get more granular in the labels, and go beyond the 5 most common classes. It is more realistic to believe that there are indeed more than 5 major classes of galaxies in the universe.

## Part B. Unsupervised Learning

### *Methods Description*

Our unsupervised learning looked at how effectively machine learning can discover galactic structural families without human intervention. For the labeled dataset, we applied the K-Means algorithm directly to the morphological data, using the same features we used in the unsupervised portion. By withholding the class labels during the clustering process, this approach acts as an independent test to determine if standard geometric properties naturally group galaxies into distinct structural families outlined by the Hubble Tuning Fork diagram.

For the unlabeled JWST dataset, we applied an unsupervised CNN and a DBSCAN model. Since space-based image profiles have varying intensities, we selected the features of flux, which were the images, and ellipticity. Initial iterations using default hyperparameter configurations revealed a critical limitation: the data was so tightly clustered around the baseline feature space that everything collapsed into a single cluster. To resolve this, we optimized the density boundaries by decreasing the epsilon parameter and increased the minimum sample value. This forced the model to isolate highly dense core regions from peripheral cosmic noise.

To accommodate the additional filters used by the JWST, we adapted a standard ResNet18 architecture to process a 7-channel multi-array band. The model was trained end-to-end to minimize the mean squared reconstruction error (MSE).

### Unsupervised Evaluation

To evaluate the unsupervised models, we used silhouette scores as our primary metric. Because a silhouette score is a measure of the compactness of a cluster, it provides an objective measure of structural boundaries. This sets us up better for evaluations whether unlabeled space imaging naturally aggregates into discrete families. We hypothesized galaxies of similar shape would form a cluster and result in higher silhouette scores. This was not necessarily the case.

When running the KMeans clustering on the GalaxyML dataset, the number of clusters that resulted in the highest silhouette score was  $k=2$ , with a weak score of 0.2776. This indicates that hand-engineered tabular shapes possess highly diffuse, overlapping boundaries that fail to naturally delineate clean cosmic families without explicit supervision.

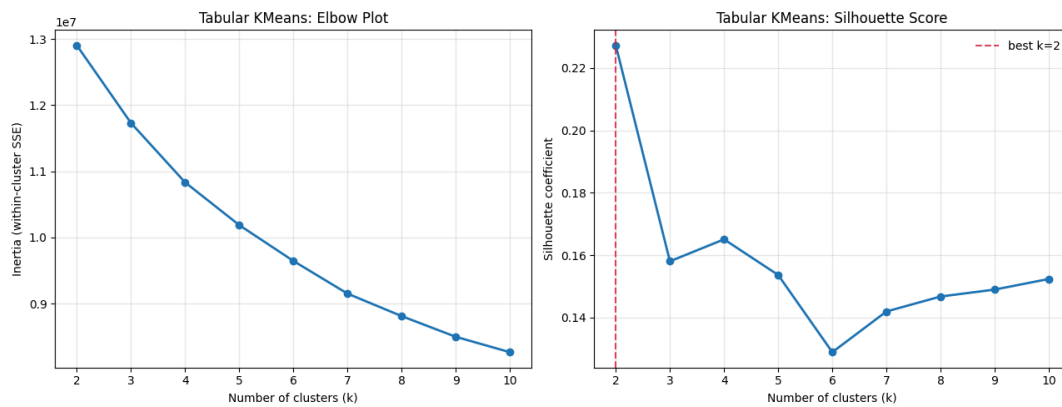


Figure 11: Left—Kmeans inertia v number of clusters  
Right—Silhouette scores across cluster counts  $k = 2 - 10$ , showing  $k = 2$  yields the highest silhouette score.

Following hyperparameter tuning, the DBSCAN model isolated one dominant cluster with two minor clusters and a cloud of outliers (noise). This indicates the DBSCAN is not finding strong multi-cluster structure in the features used.

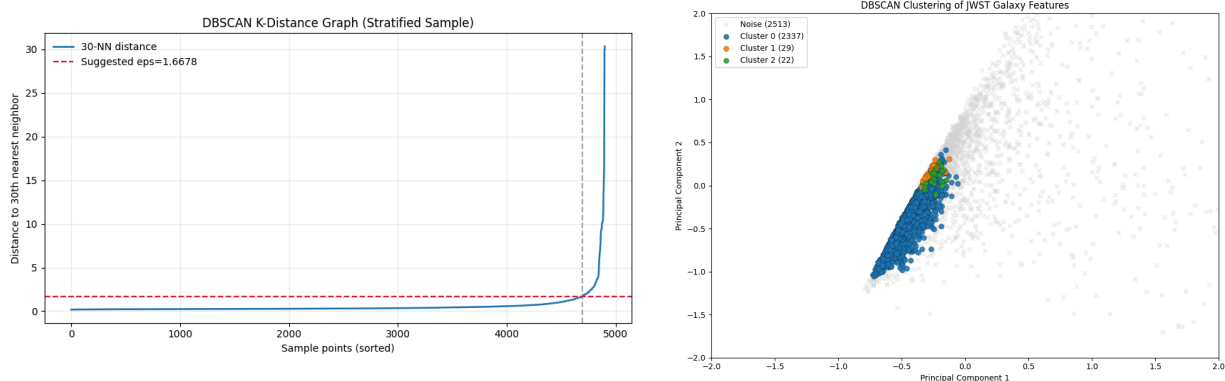
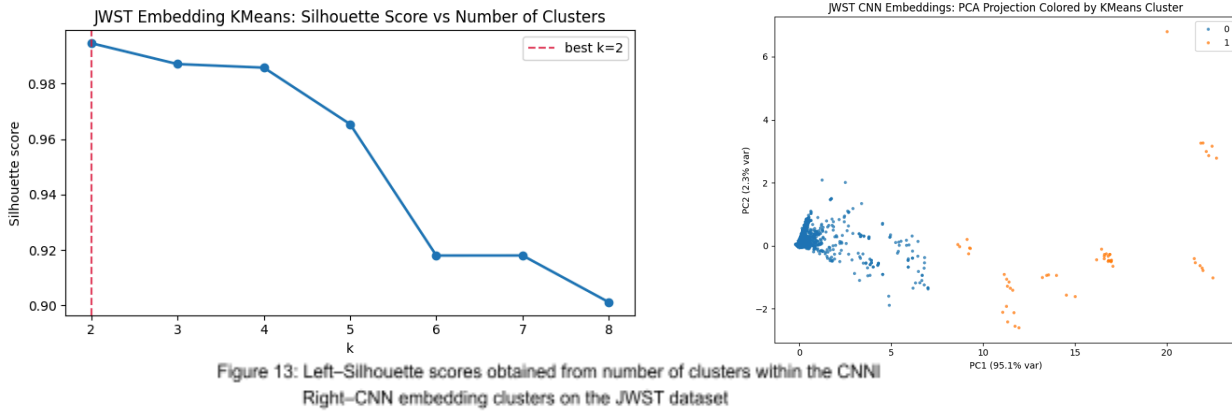


Figure 12: Left—DBSCAN K-Distance graph to find best epsilon for hyperparameter tuning.  
Right—Clusters generated from the DBSCAN model run on JWST galaxy features.

In contrast to the tabular methods, running clustering algorithms on the image features yielded good separation. At  $k=2$ , the deep-learned features achieved a Silhouette score of 0.9937. This performance gap reveals a critical insight: raw multi-band image pixels processed through deep convolutional layers preserve clear, geometric configurations that the tabular metrics can't hold.

The unsupervised CNN run on the unlabeled JWST dataset resulted in much higher silhouette scores, the highest also being at  $k=2$ , resulting in a score of .9937.



### Sensitivity analysis

Our two original unsupervised learning models we ran were KMeans clustering and DBSCAN, with the DBSCAN model performing significantly better with much higher average silhouette score. The hyperparameter tuning performed during the DBSCAN led us naturally to sensitivity analysis, even though that was not our best performing model. The silhouette values range from -0.5522 to 0.5259 when tuning hyperparameters `min_samples`, `n_clusters`, and `noise_fraction`. The DBSCAN model on the JWST data is very sensitive to hyperparameters, especially with `n_clusters` produced being highly affected by `eps`. After running these two models, we explored an unsupervised CNN. Since running the CNN takes a significant amount of computing power and time, we didn't run a sensitivity analysis on the model. Given more time, we would explore our findings with the unsupervised CNN model more.

| eps  | min_samples | n_clusters | noise_fraction | silhouette    |
|------|-------------|------------|----------------|---------------|
| 0.15 | 10          | 30         | 0.7337         | -0.5522730408 |
| 0.15 | 20          | 7          | 0.81578        | -0.1676382098 |
| 0.15 | 30          | 5          | 0.85098        | 0.415626862   |
| 0.15 | 40          | 5          | 0.88358        | 0.5033470317  |
| 0.2  | 10          | 17         | 0.48978        | -0.6620873616 |
| 0.2  | 20          | 3          | 0.56986        | -0.368703726  |
| 0.2  | 30          | 6          | 0.6285         | -0.344223612  |
| 0.2  | 40          | 3          | 0.67302        | 0.2350818284  |
| 0.25 | 10          | 7          | 0.32638        | -0.5297289553 |
| 0.25 | 20          | 1          | 0.38562        |               |
| 0.25 | 30          | 1          | 0.42788        |               |

|      |    |   |         |              |
|------|----|---|---------|--------------|
| 0.25 | 40 | 1 | 0.46046 |              |
| 0.3  | 10 | 2 | 0.22596 | 0.5220921553 |
| 0.3  | 20 | 1 | 0.26782 |              |
| 0.3  | 30 | 1 | 0.2945  |              |
| 0.3  | 40 | 1 | 0.31718 |              |
| 0.35 | 10 | 1 | 0.16024 |              |
| 0.35 | 20 | 2 | 0.1912  | 0.5259810594 |
| 0.35 | 30 | 1 | 0.21196 |              |
| 0.35 | 40 | 1 | 0.22852 |              |

Figure 14: Sensitivity analysis table of DBSCAN model

## Discussion

### Part A Learnings

The execution of the supervised learning section showed that manually choosing the morphological features without image data resulted in lower performances. Both logistic regression and random forest mainly used morphological features. While easily interpretable, our results show that these models fail to capture the same contextual information provided by image data.

In contrast to the logistic regression and random forest model, the CNN used the raw pixels from the images that resulted in a much higher F1 score of .79 and higher minority class recall at .61. The main takeaway from part A is that image data provides superior detailed information compared to tabular morphological data, especially when dealing with complex galactical structures.

### Part B Learnings

The unsupervised learning methods that we pursued were unsupervised CNN, KMeans and DBSCAN. We reached the best silhouette score of 0.9 from our unsupervised CNN when  $k=2$ . From our KMeans model, we were able to use the generated clusters as features back in our original supervised models. This resulted in relatively similar performance for the random forest model and logistic regression.

Results from our DBSCAN model showed that the data points were distributed very closely together, making it hard to generate multiple clusters without making adjustments to the model's hyperparameters. With the exception of outliers, these results indicate that most galaxies are relatively similar, having feature values that fall close together. To balance time and processing power constraints, we only ran our model on one of the smaller data files (Ngdeep, a survey dataset) under the JWST datasets. However, the Ngdeep dataset contains almost 5,000 data points which was still sufficient for our model. If given more time, we would run our model on multiple larger data sets including other survey datasets as well.

Surprisingly we got fewer clusters with JWST data since the images are directly from space and therefore had less noise than the GalaxyML dataset, since the GalaxyML images are obtained by having to go through the atmosphere.

## Ethical Considerations

When it comes to ethical consideration, it should be noted that there is no personal data or potentially harmful uses of our data analysis.

Due to the disparity in the class sizes, it is possible that there could be some bias in the supervised classification models. If any astronomers or other researchers were to rely on this model, they should be aware of the risk of a systemic bias towards the more common galaxy types. Any following classifications that utilize the models created in this project should share a disclaimer of the potentially skewed outputs, as they may exclude rare or unknown galaxy types.

Not meant to replace citizen astronomers or people, we want to provide additional tools utilizing unsupervised learning due to the sheer size of datasets. These should be used sparingly as tools to aid humans in galaxy dataset processing and evaluation.

## Statement of Work

| Nancy Melchert                                            | Sophia Menchaca                           | Blythe Weng                          |
|-----------------------------------------------------------|-------------------------------------------|--------------------------------------|
| Professional Video Editing<br>(Galaxy and Cat animations) | Coding - random forest, kmeans clustering | Coding - logistic regression, DBSCAN |
| Coding - supervised and unsupervised CNN                  | Report writeup                            | Report writeup                       |
| Report writeup                                            | Standup                                   | Standup                              |
| Standup                                                   |                                           |                                      |

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